Luckie Musngi

(520) 999-5482 | luckie.musngi@gmail.com | https://github.com/LuckieMusngi/

EDUCATION

University of Arizona (Expected May 2026) **Tucson, AZ**

- **Major:** Computer Science BS, **Minor:** Mathematics
- **GPA:** 3.5
- **Relevant Coursework:** Data Science, OOP, Discrete Math, Differential Equations, Linear Algebra, Probability

TECHNICAL SKILLS AND

Programming: C#, Python, Java, Javascript, HTML/CSS, C, MIPS, C++, OpenSCAD

Frameworks / Tools: React.js, Unity, NumPy, Pandas, Matplotlib, Scikit-learn, Git/GitHub, REST APIs

Other skills: Agile Scrum, Embedded Systems, Hardware–Software Integration, Rapid Prototyping, Real-Time Control, Sensors, Servo/DC Motors, 3D Modeling, Mechanical Design

WORK EXPERIENCE

Vail Academy and High School **Tucson, AZ**

Site Tech Coordinator Shadow 09/2021-11/2021

- Gained hands-on experience troubleshooting hardware, software, and network issues; configured technical equipment for events and solved day-to-day problems.
- Learned about IT support workflows, including ticketing systems, troubleshooting protocols, and general problem-solving while collaborating with staff and faculty.

Lins Grand Buffet **Tucson, AZ**

Expeditor 07/2022-Present

- Collaborated with the team to maintain stocked and presentable areas while coordinating specialists for efficient buffet operations.
- Managed and resolved customer requests, communicating with multiple teams in a fast-paced environment.

PROJECTS

High-Speed Arduino Sorting Machine | https://github.com/LuckieMusngi/Arduino-Skittle-Sorter_Quannel-Method

- **World Record!** (~4.44 skittles/sec)
- Designed and built solo, a fully custom, high-throughput color-based sorting system using Arduino.
- Engineered a four-way funnel and circular trapdoor mechanism enabling parallel sorting. Modeled and 3D-printed all components in OpenSCAD.
- Integrated multiple RGB sensors, servo motors, and a geared DC motor for real-time classification and routing. Optimized timing, throughput, and mechanical–software coordination.

Arduino, Embedded C/C++, OpenSCAD, Sensors, Servo/DC Motors, Rapid Prototyping, Real-Time Control, Parallel Processing

Mock Ski Resort SQL Database System | https://github.com/LuckieMusngi/Ski-Resort-Database

- Co-developed (2-person) a full Oracle SQL database system with Java JDBC interface.
- Planned and designed relational schema, constraints, and required queries given an application domain.
- Built a command-line interface to query and modify the database (add/update/delete operations), and implemented all table creation and data population scripts, handling all debugging and validation.

Oracle SQL, Java, JDBC, Relational Schema Design, E-R Diagrams, DDL/DML, CLI, Database Constraints

Crime in Tucson Analysis | https://github.com/LuckieMusngi/Crime-in-Tucson-Analysis

- Analyzed official Tucson crime data to identify patterns and predict crime levels.
- Preprocessed and cleaned data using NumPy and Pandas.
- Applied linear models and classification techniques; evaluated models using cross-validation and confusion matrices, and visualized trends and model outputs with Matplotlib.

Python, Jupyter Notebook, NumPy, Pandas, Matplotlib, Scikit-learn, Linear Algebra, Probability, Machine Learning, Data Preprocessing, Feature Engineering, Model Evaluation

WeatherApp | https://github.com/LuckieMusngi/Weather

- Built a responsive weather web application using React.js.
- Integrated the OpenWeatherMap API to retrieve real-time weather for user-input cities.
- Implemented interactive UI components and handled asynchronous API calls.

JavaScript, HTML/CSS, React.js, REST APIs, Responsive UI, Asynchronous Programming

CLUBS AND ORGANIZATIONS

Software Engineering Wildcats	Member	08/2022-Present
UA Climbing Comp Team	Member	10/2024-Present